MAT211: Calculus-II

Problem Sheet #03 **Topic: Partial Derivatives, Directional Derivatives & Chain Rule**

Date: March 18, 2026

Problem 1 [Marks: 4]

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x, y) = x^2 y + e^{3x+y}.$$

Find all second-order partial derivatives of f.

Problem 2 [Marks: 6]

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} \dfrac{xy(x^2 - y^2)}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Calculate the mixed partial derivatives

$$f_{xy}(0, 0) = \frac{\partial^2 f}{\partial x \partial y}(0, 0), \qquad f_{yx}(0, 0) = \frac{\partial^2 f}{\partial y \partial x}(0, 0),$$

and show that

$$f_{xy}(0, 0) \neq f_{yx}(0, 0).$$

Problem 3 [Marks: 4]

If $T(\mathbf{x})$ is a linear transformation, find the directional derivative of $T(\mathbf{x})$ at $\mathbf{x} = \mathbf{a}$ in the direction $\mathbf{v}$.

Problem 4 [Marks: 4]

If $f : \mathbb{R}^n \to \mathbb{R}$ is defined by

$$f(\mathbf{x}) = \|\mathbf{x}\|^2,$$

find the directional derivative of f at $\mathbf{x} = \mathbf{a}$ in the direction $\mathbf{v}$.

Problem 5 [Marks: 6]

Let

$$f(x, y) = \begin{cases} \dfrac{x^3}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Find the partial derivatives and all directional derivatives of f at $(0, 0)$.

Problem 6 [Marks: 4]

Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x, y) = x^2 - y^2.$$

Find the derivative of f at the point

$$\begin{pmatrix} 2 \\ 3 \end{pmatrix},$$

and hence determine the value of

$$Df\left(\begin{pmatrix} 2 \\ 3 \end{pmatrix}\right).$$

Problem 7 **[Marks: 4]**

For $f : \mathbb{R}^2 \to \mathbb{R}^2$ given by

$$f(x, y) = \begin{pmatrix} xy \\ y^2 \end{pmatrix},$$

find the value of

$$Df\left(\begin{pmatrix} a \\ b \end{pmatrix}\right).$$

Problem 8 **[Marks: 4]**

Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by

$$f\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} xy \\ x^2 - y^2 \end{pmatrix}.$$

At the point $\mathbf{a} = (1, 1)$, at what rate is f varying in the direction $\mathbf{v} = (1, 0)$?

Problem 9 **[Marks: 5]**

The (x, y)-coordinates of a roller coaster at time t are given by

$$(x(t), y(t)) = (\cos t, \sin t),$$

and its height at (x, y) is given by

$$z = x^2 y + x + 1.$$

Find its rate of change in height with respect to time at time t.

Problem 10 **[Marks: 5]**

Given that

$$F : \mathbb{R} \to \mathbb{R}^3, \qquad F(t) = \begin{pmatrix} t^2 \\ t^4 \\ -t^3 \end{pmatrix},$$

and

$$G : \mathbb{R}^3 \to \mathbb{R}, \qquad G\left(\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}\right) = \sin(3x_1 + 2x_2 + 5x_3).$$

Find

$$\left.\frac{d}{dt}(G \circ F)\right|_{t=1}.$$

Problem 11 **[Marks: 5]**

Given that

$$F : \mathbb{R}^2 \to \mathbb{R}^3, \qquad F\left(\begin{pmatrix} u \\ v \end{pmatrix}\right) = \begin{pmatrix} u+v \\ u^2 \\ v^3 \end{pmatrix},$$

and

$$G : \mathbb{R}^3 \to \mathbb{R}, \qquad G\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = xy + z.$$

Find all partial derivatives of $G \circ F$ at $(u, v) = (1, 1)$.

Problem 12 **[Marks: 6]**

Given that

$$F : \mathbb{R}^2 \to \mathbb{R}^2, \qquad F\left(\begin{pmatrix} u \\ v \end{pmatrix}\right) = \begin{pmatrix} u^2+v \\ uv \end{pmatrix},$$

and

$$G : \mathbb{R}^2 \to \mathbb{R}^2, \qquad G\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x+y \\ x-y \end{pmatrix}.$$

Find

$$D(G \circ F)|_{\mathbf{x}=(1,1)^T}.$$

Then verify that the partial derivatives with respect to u and v of the component functions agree with the corresponding entries of the matrix $D(G \circ F)$.

Problem 13 **[Marks: 5]**

Suppose the temperature in space is given by

$$f\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = xyz^2 + e^{3xy - z^2},$$

and the position of a bumblebee is given as a function of time by a differentiable map

$$g : \mathbb{R} \to \mathbb{R}^3.$$

If at time $t = 0$ the bumblebee is at

$$a = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

and her velocity vector is

$$\mathbf{v} = \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix},$$

find the rate at which she perceives the temperature to be changing at that instant.